

Lecture 7

Solving Square Linear System II: Breakdown Cases and General Solving

Instructors:

L01 Cosme Louart 程高木

L02 Junfeng Wu 吴均峰

L03&L04 Ruoyu Sun 孙若愚



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

数据科学学院

School of Data Science

Comments on Course Contents Arrangement

Theme of Lec 3-15: Solving Linear Systems

(along with Theory of Matrix & Linear Space)

Why learning Matrix?

It will be a fundamental tool of AI & data science

Why learning linear systems?

- A very fundamental problem, but you may or may not directly encounter it
- Purpose of learning it:
 - half for its usefulness,
 - half for motivating matrix & linear space theory

Course arrangement:

We spent Lec 3 & 4 on linear system solving (preliminary).

Then spent Lec 5 & 6 on matrix algebra (multiplication & partition).

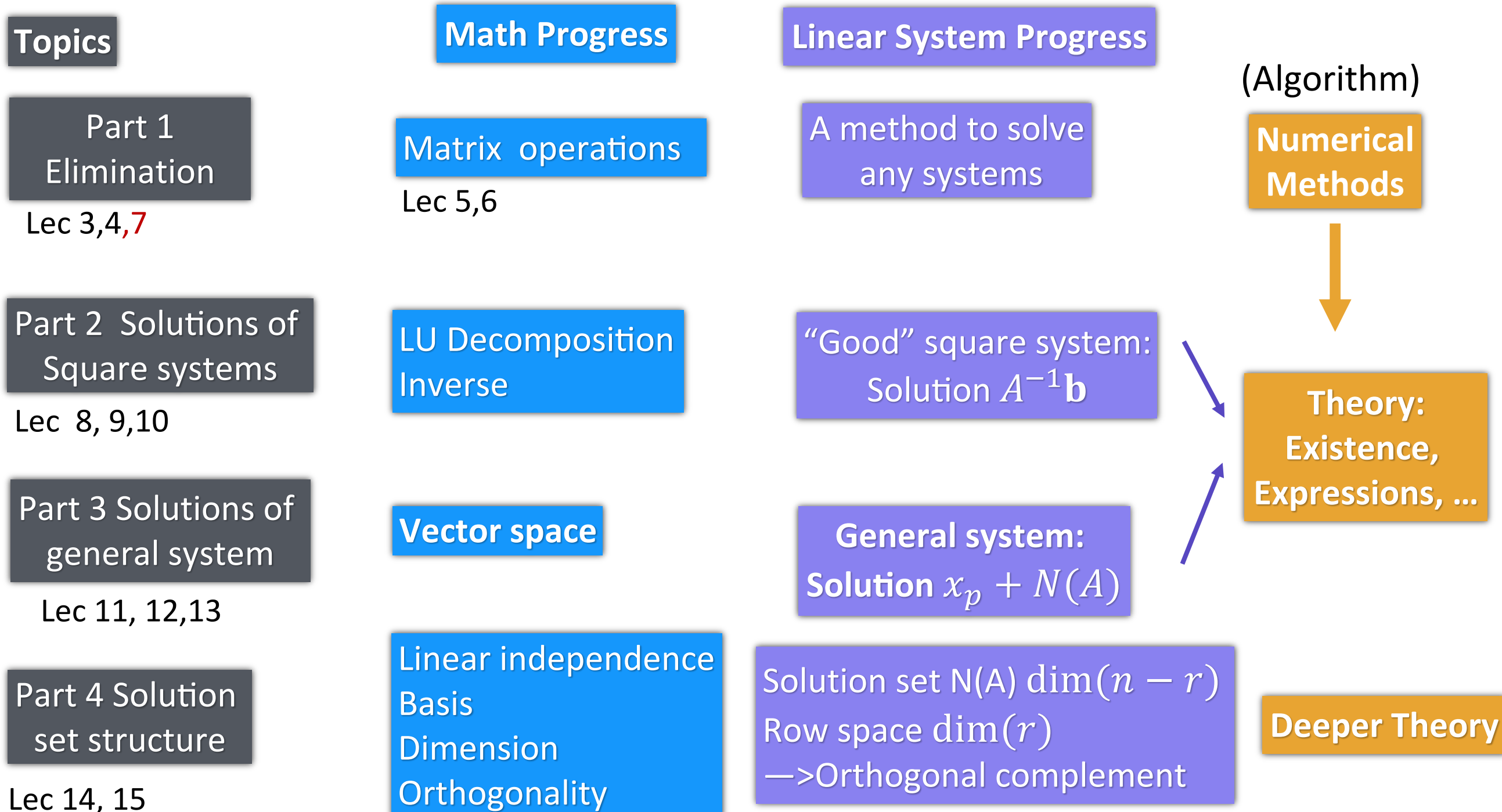
--as preparation for future study

From Lec 7: come back to solving linear systems

Roadmap: *Topics Before Midterm*

Lec 1,2: Vector operations.

Theme of Lec 3-15: Solving Linear Systems (along with Theory of Matrix & Linear Space)



Outline & Learning Goals

Today ... Strang's book v5: Sec 2.2, Sec. 3.2

1. Breakdown Cases of Solving Square Linear System
2. REF, RREF

After this lecture, you should be able to

1. Handle a few breakdown cases for solving square linear system
2. Explain what are REF and RREF
3. Memorize that GJ-E for any square system will produce upper triangular matrix and RREF

Part 0 Preview

5 mins

Is 2*2 Linear System Easy?

Primary school:

$$\begin{cases} x_1 + 2x_2 = 3, \\ 2x_1 - 3x_4 = 4. \end{cases} \quad \begin{cases} 3x_1 + 2x_2 = 1, \\ 7x_1 - 5x_4 = 4. \end{cases} \quad \begin{cases} x_1 - 2x_2 = 3, \\ 4x_1 + 7x_4 = -9 \end{cases} \quad \dots$$

Is the university course about solving bigger systems?

Solve a 2*2 linear system? So easy!

Really?



Is 2*2 Linear System Easy?

Primary school:

$$\begin{cases} x_1 + 2x_2 = 3, \\ 2x_1 - 3x_4 = 4. \end{cases} \quad \begin{cases} 3x_1 + 2x_2 = 1, \\ 7x_1 - 5x_4 = 4. \end{cases} \quad \begin{cases} x_1 - 2x_2 = 3, \\ 4x_1 + 7x_4 = -9 \end{cases} \quad \dots$$

Is the university course about solving bigger systems?
Solve a 2*2 linear system? So easy!



Really?

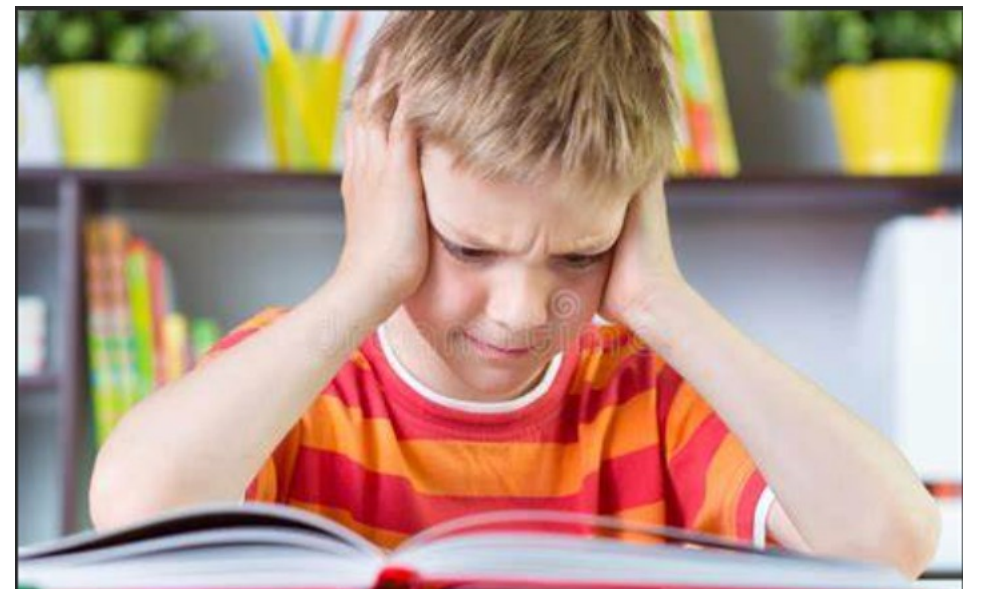
University:

Check this one:

$$\begin{cases} -x_1 + x_2 = 3, \textcircled{1} \\ 2x_1 - 2x_2 = 4. \textcircled{2} \end{cases}$$

$\textcircled{2} - \textcircled{1} \times 2$, get: $0 = 5$??

The solution of the system is nothing.



Unknown Unknowns

There are known knowns; there are things we know that we know.

There are known unknowns; that is to say, there are things that we now know we don't know.

But there are also unknown unknowns – there are things we do not know we don't know.

-Donald Rumsfeld



Knowns Unknowns	Known Knowns Things we are aware of and understand.	Known Unknowns Things we are aware of but don't understand.
	Unknown Knowns Things we understand but are not aware of.	Unknown Unknowns Things we are neither aware of nor understand.
	Knowns	Unknowns

Unknown unknowns (for most, if not all, 5th grade students):

Ask: Do you know how to solve **all** 2×2 linear systems?

5th grade student: yes, I know. [**they thought they know**]

But actually... no [**they don't know that they don't know**].

Solving all 2×2 system in primary schools does NOT Mean solving ALL 2×2 system

Most 5th grade students don't even know what “solve” really means:

It means “**finding solution set**” (explain later)!

Part I “Breakdown” of Solving Square Systems

Strang's book v5, Sec 2.2, part “Breakdown of elimination”

25 mins

Pipeline of GJ-E for Solving (good) Square Systems

Phase I: Forward elimination (make zeros *below* pivots)

$$\begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{bmatrix} \rightarrow \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{bmatrix} \rightarrow \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & \times & \times \end{bmatrix} \rightarrow \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{bmatrix}$$

Upper triangular matrix

Phase II: Backward elimination.

(a) Make pivots = 1 (optional but standard before clearing above):

$$\begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \times & \times & \times \\ 0 & 1 & \times & \times \\ 0 & 0 & 1 & \times \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) Make zeros above pivots; column by column.

$$\begin{bmatrix} 1 & \times & \times & \times \\ 0 & 1 & \times & \times \\ 0 & 0 & 1 & \times \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \times & \times & 0 \\ 0 & 1 & \times & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \times & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Diagonal matrix

Summary of GJ-E for Solving Square Systems

Phase 1: Forward Elimination.

Perform elementary row operations and try to get an **upper triangular** matrix.

Phase 2: Backward substitution

Perform elementary row operations and try to get a **diagonal matrix**.

Assumption 1

At each iteration of the forward elimination, the next diagonal entry is nonzero.

Claim 1 Under Assumption 1, we can get a diagonal matrix at the end of Step 2.

Corollary 1 Under Assumption 1, the system has a unique solution.

This assumption may not hold for some problems; will discuss later.

Pivot

Example

$$\begin{cases} x_1 + x_2 = 2, \\ 2x_1 - 2x_2 = 4. \end{cases}$$

Subtract 2* Eq.1 from Eq. 2

Matrix form of GJ-E (1st step):

pivot $\rightarrow$ $\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & -2 & 4 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -4 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right]$

Pivot = first nonzero in the row that does the elimination

The pivots are on the diagonal of the triangle after elimination

Pivot: Examples

What are the pivots in the process?

1st pivot

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 2 & 9 \\ 1 & 2 & 3 & 16 \end{array} \right]$$

Apply $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & 2 & 4 \end{array} \right]$$

*check 2nd row
1st nonzero entry as 2nd pivot*

Apply $R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

*check 3rd row
1st nonzero entry as 3rd pivot.*

Question: What if there are zeros? Type 1

Example 1

[no solution, < 2 pivots]
Find (x_1, x_2) s.t. the following equations hold simultaneously.

$$\begin{cases} x_1 + x_2 = 2, \\ 2x_1 + 2x_2 = 5. \end{cases}$$

Subtract $2 \times$ Eq. 1 from Eq. 2

Slightly different from the previous example

Matrix form of GJ-E:

1 pivot(s) in total

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 5 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right]$$

Find (x_1, x_2) s.t. equations hold simultaneously

$$\begin{cases} x_1 + x_2 = 2 & \textcircled{1} \\ 0 = 1 & \textcircled{3} \end{cases}$$

$$\begin{aligned} \text{Find } (x_1, x_2) \text{ s.t.} \\ x_1 + x_2 = 2 \\ \text{and } 0 = 1. \end{aligned}$$

Zero is never allowed as a pivot! No second pivot.

Impossible to find

Conclusion: nothing is a solution of the system.

Question: What if there are zeros? Type 2

Example 2 [∞ -many solutions; < 2 pivots.]

$$\begin{cases} x_1 + x_2 = 2, \\ 2x_1 + 2x_2 = 4. \end{cases} \quad \begin{cases} 1+1=2 \\ 2 \cdot 1 + 2 \cdot 1 = 4 \end{cases}, \quad \begin{cases} 3-1=2 \\ 2 \cdot 3 + 2 \cdot (-1) = 4 \end{cases}$$

Subtract $2 \times$ Eq. 1 from Eq. 2

Slightly different from the previous example

Matrix form of GJ-E:

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 4 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right]$$

1 pivot(s) in total

$$\text{Find } (x_1, x_2), \text{ s.t. } \begin{cases} x_1 + x_2 = 2 \\ 0 = 0 \end{cases}, \text{ i.e.}$$

find (x_1, x_2) s.t.
 $x_1 + x_2 = 2$
 ~~$0 = 0$~~
useless

Conclusions:

(a) $(t, 2-t), \forall t \in \mathbb{R}$ is a solution of the system.

(b) The system has infinitely many solution(s).
 ∞ -many

Question: What if there are zeros? Type 3

Example 3 [Temporary failure (zero in diagonal).
row-exchange operation can produce 2 pivot(s) .]

$$\begin{cases} x_2 = 2, \\ 2x_1 + 2x_2 = 4. \end{cases}$$

Matrix form of GJ-E:

$$\left[\begin{array}{cc|c} \boxed{0} & 1 & 2 \\ 2 & 2 & 4 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} \boxed{2} & 2 & 4 \\ 0 & \boxed{1} & 2 \end{array} \right]$$

0 pivot
(seems)

2 pivot(s)

Unique solution.

Visualization of Various Cases: 2 by 2 System

- Visualization of a 2×2 system of linear equations

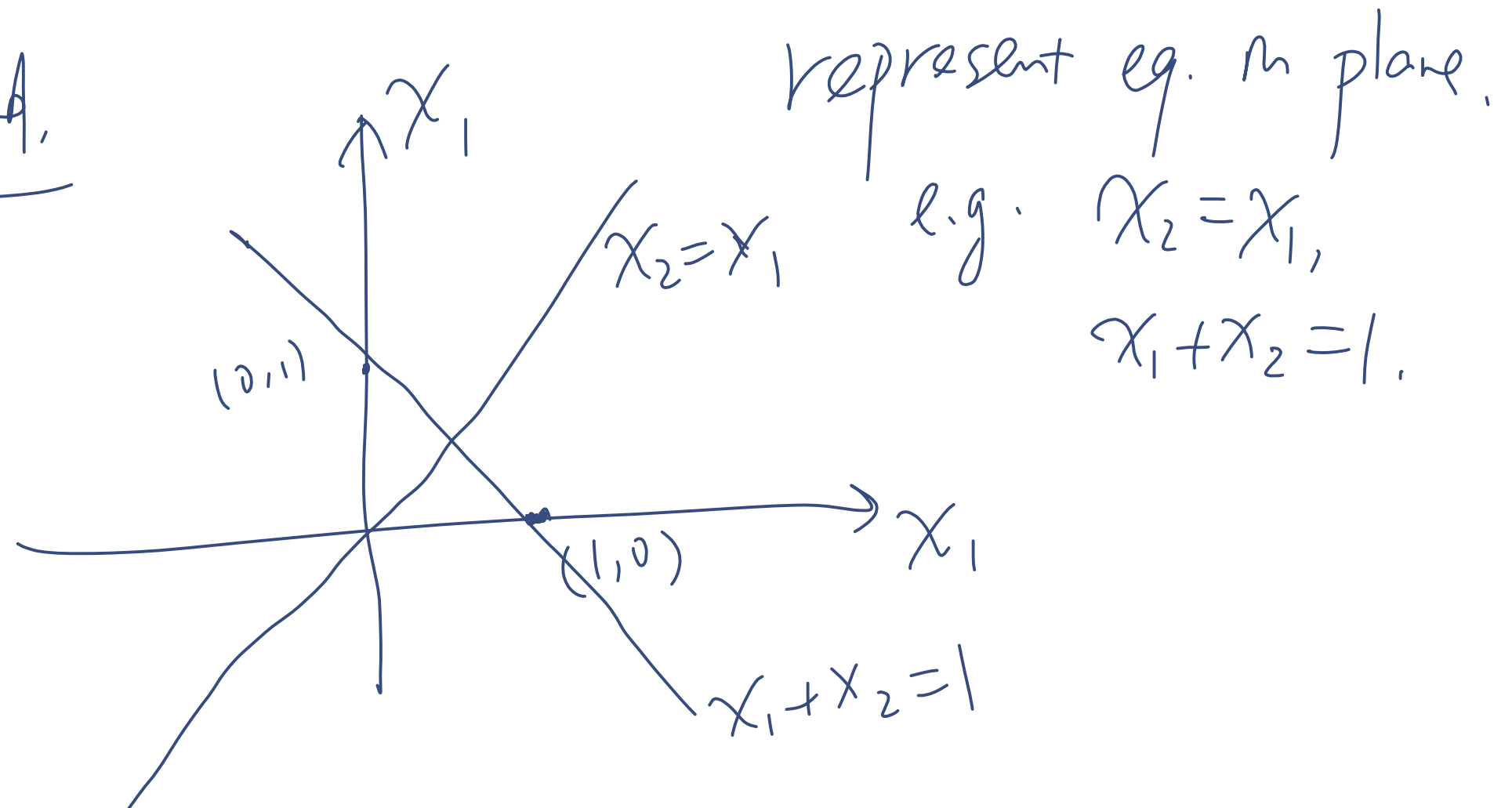
$$(i) \begin{cases} x_1 + x_2 = 2 \\ x_1 - x_2 = 2 \end{cases}$$

$$(ii) \begin{cases} x_1 + x_2 = 2 \\ x_1 + x_2 = 1 \end{cases}$$

$$(iii) \begin{cases} x_1 + x_2 = 2 \\ -x_1 - x_2 = -2 \end{cases}$$

Draw the lines for these equations. What do you observe?

Background.



Visualization of Various Cases: 2 by 2 System

- Visualization of a 2×2 system of linear equations

(i) $x_1 + x_2 = 2$

$x_1 - x_2 = 2$

unique soln

(ii) $x_1 + x_2 = 2$

$x_1 + x_2 = 1$

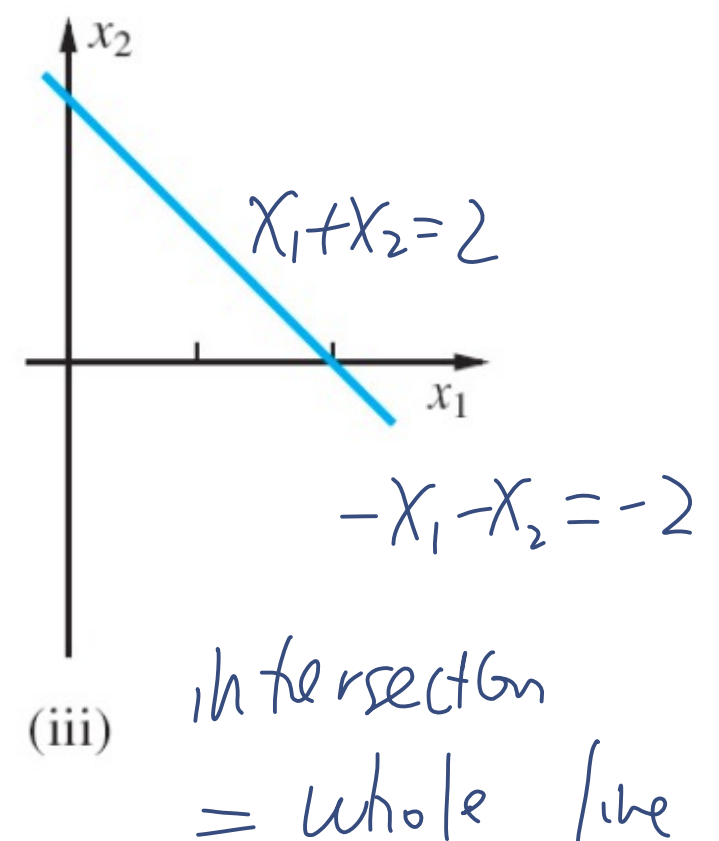
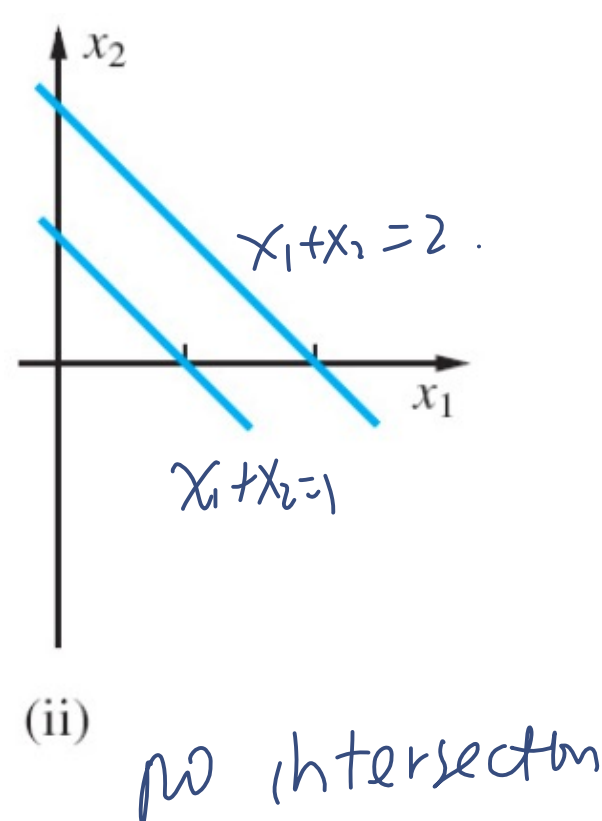
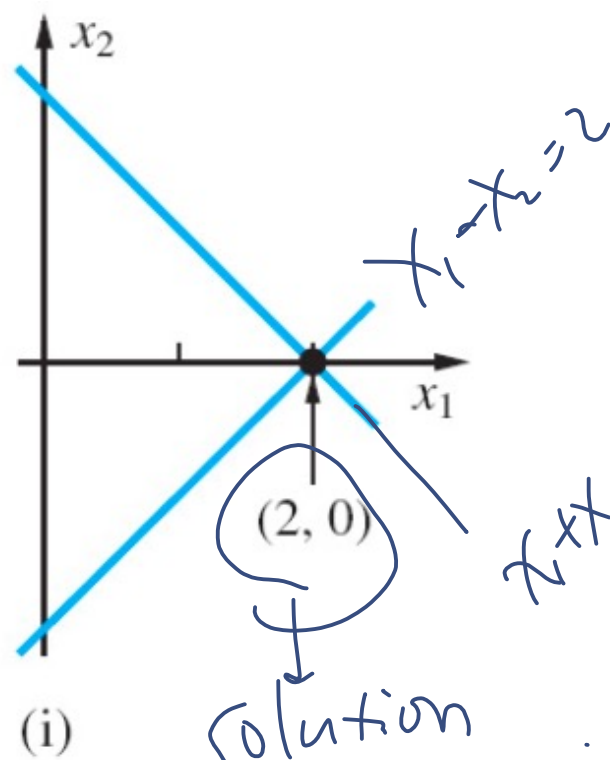
no solution

(iii) $x_1 + x_2 = 2$

$-x_1 - x_2 = -2$

∞ -many solutions

Draw the lines for these equations. What do you observe?

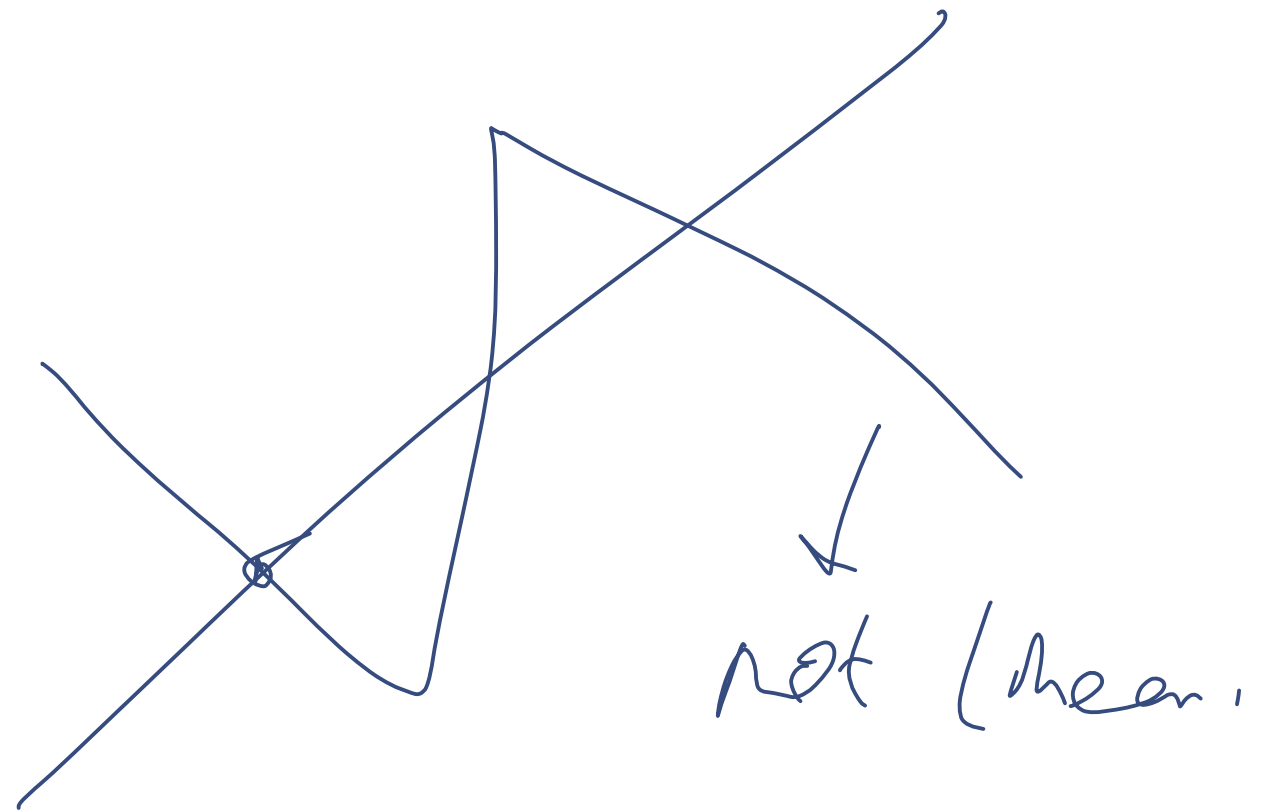
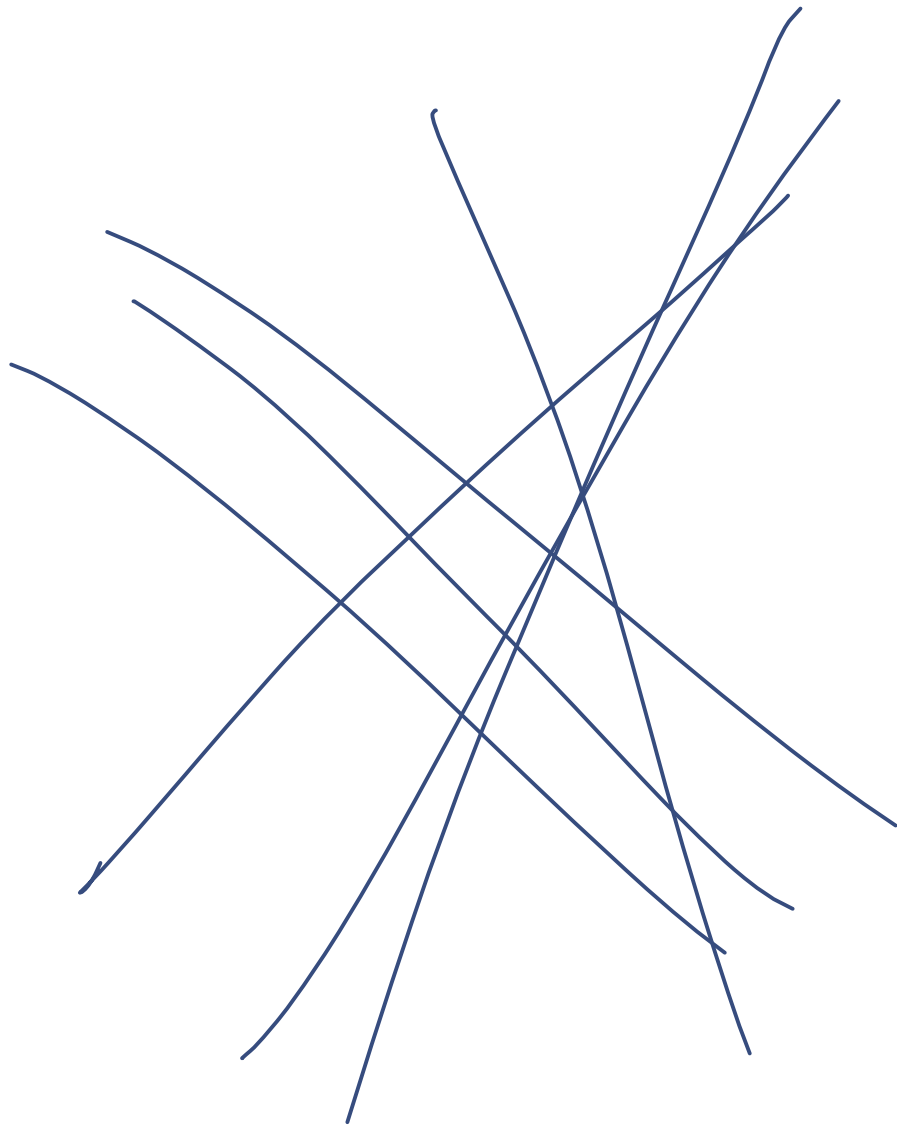


Judgement Question

Judgement: (True or False)

A linear system can have exactly 3 solutions.

False.



Solution Set

Definition (Solution and Solution Set)

A **solution** of an $m \times n$ linear system with variables $(x_1, \dots, x_n)$ is a vector $(s_1, \dots, s_n)$ such that if we let $x_i = s_i$ for all $i = 1, \dots, n$ the m equations hold simultaneously.

The **solution set** of an $m \times n$ linear system is a set that contains all solution(s).

$$\emptyset, \quad \{\vec{0}\}, \quad \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots\}$$

Examples of Solution Sets

Example (Solution Set)

$$\begin{aligned} 2x_1 + 3x_2 &= 3, \\ x_1 - x_2 &= 4 \end{aligned} \quad (*)$$

The solution set is $\{(3, -1)\}$ Pay attention to the double bracket.

$(3, -1)$ is the unique solution of $(*)$.

$\Leftrightarrow$ The solution set of $(*)$ is $\{(3, -1)\}$

Examples of Solution Sets

Example (Solution Set)

$$2x_1 + 3x_2 = 3,$$

$$4x_1 + 6x_2 = 6$$

$$\left[\begin{array}{cc|c} 2 & 3 & 3 \\ 4 & 6 & 6 \end{array} \right] \xrightarrow{R_2 \cdot \frac{1}{2}} \left[\begin{array}{cc|c} 2 & 3 & 3 \\ 2 & 3 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{cc|c} 2 & 3 & 3 \\ 0 & 0 & 0 \end{array} \right]$$
$$\xrightarrow{R_1 \cdot \frac{1}{3}} \left[\begin{array}{cc|c} \frac{2}{3} & 1 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$\left\{ \frac{2}{3}x_1 + \right.$$

$$x_2 = 1$$

$$0 = 0$$

i.e.

i.e.

$$x_2 = 1 - \frac{2}{3}x_1 \text{ is a solution}$$

$$\left(t, \frac{3-2t}{3} \right) \text{ is a solution, } \forall t.$$

The solution set is $\left\{ \left(t, \frac{3-2t}{3} \right) \mid t \in \mathbb{R} \right\}$

(Infinitely many)

Will learn how to derive this set in later lectures

Part II GJ-E and RREF

Strang's book v5, Sec 2.2, part "Breakdown of elimination"

Strang's book v5, Sec 3.2, part "Reduced Row Echlon Form R"

35-40 mins

Have We Enumerated All Cases?

Q: We have seen a few breakdown cases.

Do we know how to handle **ALL** cases?

Two Simple Cases for Square Systems

Coefficient matrix A ; square matrix.

Gaussian elimination (GE) (both forward and backward)

$$A \rightarrow A_1 \rightarrow A_2 \dots \rightarrow U \rightarrow \dots B_1 \rightarrow \dots \dots \rightarrow D \quad D \text{ is diagonal?}$$

Note: we don't always get a diagonal matrix.

End of forward elimination:
Upper triangular

End of backward step

Case 1: Diagonal entries of U are nonzero.

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\rightarrow$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Diagonal matrix

Good cases

pivots

Case 2: Some diagonal entries of U are zero.

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rightarrow$

$$\begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Non-diagonal matrix.

Structure? (represent by blocks)

$$\left[\begin{array}{c|c} I & F \\ \hline 0 & 0 \end{array} \right]$$

3 pivots

What if Next Diagonal Entry is 0?

When the next diagonal entry (k,k) is 0, what to do next?

$$\begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & \textcircled{0} & & & * \\ 0 & 0 & & & & * \\ 0 & 0 & & & & * \end{bmatrix}$$

What if Next Diagonal Entry is 0?

When the next diagonal entry (k,k) is 0, what to do next?

$$\left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & & & * \\ 0 & 0 & & & & * \\ 0 & 0 & & & & * \end{array} \right]$$

Step 1: Find a non-zero entry below the next diagonal entry (k,k).

If succeed: Row exchange to make the next diagonal entry nonzero, as pivot.

If **no non-zero entry**: go to Step 2.

$$\left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & \\ 0 & 1 & * & * & * & \\ 0 & 0 & 0 & & & \\ 0 & 0 & 3 & & & \\ 0 & 0 & 2 & & & \end{array} \right] \longrightarrow \left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & \\ 0 & 1 & * & * & * & \\ 0 & 0 & 2 & & & \\ 0 & 0 & 3 & & & \\ 0 & 0 & 0 & & & \end{array} \right]$$

In this example, swap R3 and R5; use (3,3)-th entry 2 as pivot.

Remark: In this example, can swap R3 and R4 as well.

What if Next Diagonal Entry is 0? 2nd Case

Step 2: (when all entries below the next diagonal entry is 0)

Check the $(k+1)$ -th column, to find a non-zero entry as pivot
(possibly use row exchange to make sure it is in the k -th row)

In this example: Check the 4th column.

Step 2.1: If $(4,3)$ -th entry is nonzero, use it as pivot to eliminate entries below it;
If $(4,3)$ -th entry is zero, go to Step 2.2.

3rd row $\rightarrow$

$$\left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & \textcircled{1} & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \end{array} \right] \rightarrow \left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * & * \end{array} \right]$$

What if Next Diagonal Entry is 0? 3rd Case

Step 2.2: (when (4,3)-th entry is zero)

Check the 4th column; find nonzero entry 2 in 4th row.

Swap R3 and R4, to make (4,3)-th entry nonzero, and use it as pivot for elimination.

$$\begin{array}{c} \left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & 3 & * & * \end{array} \right] \end{array} \xrightarrow{\text{Swap R3 and R4}} \begin{array}{c} \left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 3 & * & * \end{array} \right] \end{array}$$
$$\xrightarrow{\text{Use pivot 2 in R3, C4}} \begin{array}{c} \left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * & * \end{array} \right] \end{array}$$

If Still All Zero, Continue...

If in Step 2, no non-zero entries in the (k+1)-th column below (k,k+1),
Then check the (k+2)-th column to find non-zero entries.

In this example: Check the 5th column.

$$\begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & \text{shaded} & * \\ 0 & 0 & 0 & 0 & \text{shaded} & * \\ 0 & 0 & 0 & 0 & \text{shaded} & * \end{bmatrix}$$

The matrix is shown with a blue vertical line between the 5th and 6th columns. The 5th column is shaded light blue. The 3rd and 4th columns have their 0 entries circled in green. The 6th column has a red asterisk in the 3rd row and black asterisks in the 4th and 5th rows. A red arrow points from the 5th column towards the right.

Continue the process.

If cannot find non-zero entry on or below k-th row in the
k-th column, (k+1)-th column, ..., n-th column,
Then everything on or below the k-th row is 0!!!

Big zero block at the bottom of the coefficient matrix.

Finish the forward process; begin backward elimination.

If Still All Zero, Continue...

If cannot find non-zero entry on or below k-th row in the k-th column, (k+1)-th column, ..., n-th column,

$$\left[\begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * \end{array} \right]$$

then everything on or below the k-th row is 0!!!

--Big zero block at the bottom.

Finish the forward process. Start backward process.

Then, use pivots to eliminate entries above.

New Case 1: Moving to Next Column For Pivot

Example 1

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 & -2 \end{bmatrix}$$

Rule:

If row exchange cannot
create a pivot,
then move to the next
column to find pivot.

New Case 1: Moving to Next Column For Pivot

Example 1

End of forward elimination:

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 & -2 \end{bmatrix} \xrightarrow{\substack{R_4 \leftarrow R_4 - R_3 \\ R_5 \leftarrow R_5 + 2R_3}} \begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & * & 0 & * \\ 0 & 1 & * & 0 & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

End of backward step

Rule:

If row exchange cannot create a pivot, then move to the next column to find pivot.

Observation:

There's a column $\begin{bmatrix} * \\ * \\ 0 \\ 0 \\ 0 \end{bmatrix}$ that stay unchanged in the rest of GJ-E.

日期: /

eliminate them by "i"

$$\begin{bmatrix} \cdot & \cdot & \cdot & 2 & 1 \\ & \cdot & \cdot & 3 & 4 \\ & 1 & & 1 & 1 \\ & & 1 & 0 & 0 \\ & 1 & & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} R_1 &\rightarrow R_1 - 2R_3 \\ R_2 &\rightarrow R_2 - 3R_3 \end{aligned}$$

$$\begin{bmatrix} \cdot & \cdot & 0 & -1 \\ & \cdot & 0 & 2 \\ & & 1 & 1 \\ & & 0 & 0 \\ & & 0 & 0 \end{bmatrix}$$

don't get
more nonzeros.

this step is useless

$$\begin{aligned} R_1 &\rightarrow R_1 + R_3 \\ R_2 &\rightarrow R_2 - 2R_3 \end{aligned}$$
$$\begin{bmatrix} \cdot & \cdot & 1 & 0 \\ & \cdot & -2 & 0 \\ & & 1 & 1 \\ & & 0 & 0 \\ & & 0 & 0 \end{bmatrix}$$

New Case 2: Moving to Next Column & Row Exchange

Example 2

Rule: For the next column: perform row exchange to get a pivot if needed.

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 & -2 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{4th row} \rightarrow} \begin{bmatrix} 1 & 0 & * & 0 & 0 \\ 0 & 1 & * & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

End of forward elimination:

End of backward step

Summarizing Examples So Far

End of forward elimination, we get these matrices:

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

End of GJ-E, we get these matrices:

$$\begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & * & 0 & 0 \\ 0 & 1 & * & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & * & 0 & 0 \\ 0 & 1 & * & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

What is the general shape?

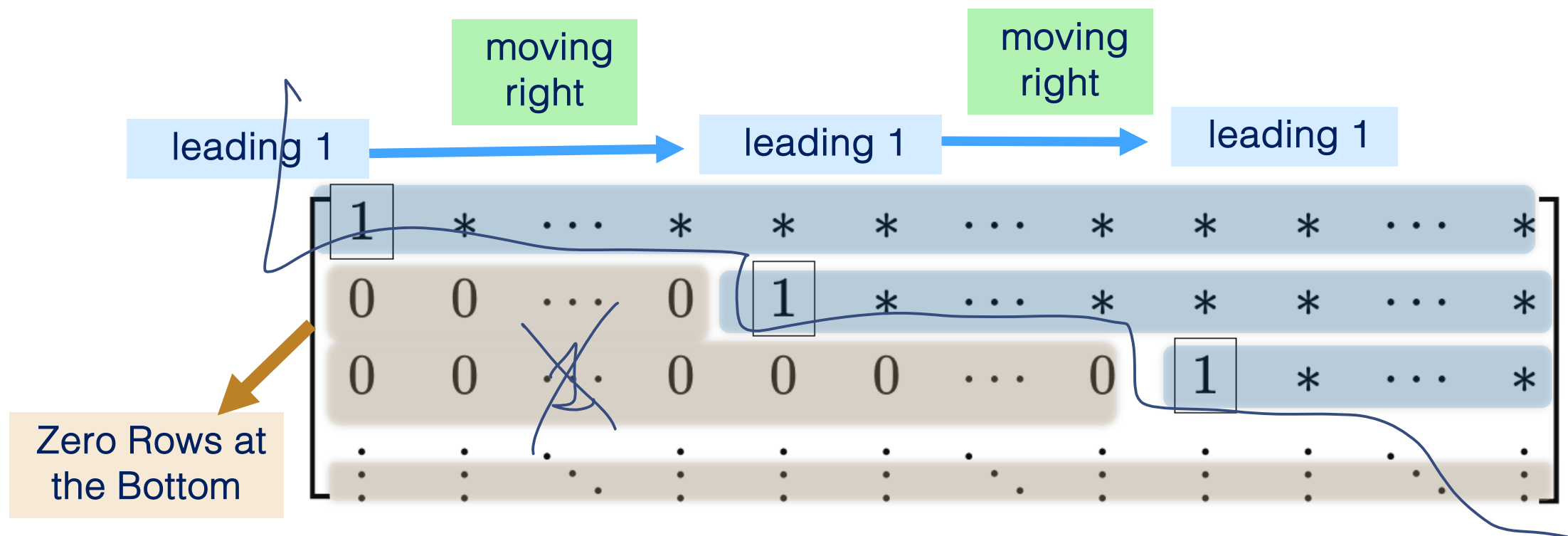
REF and Upper Triangular Matrix

Definition (Row-Echelon Form, REF) 行阶梯阵

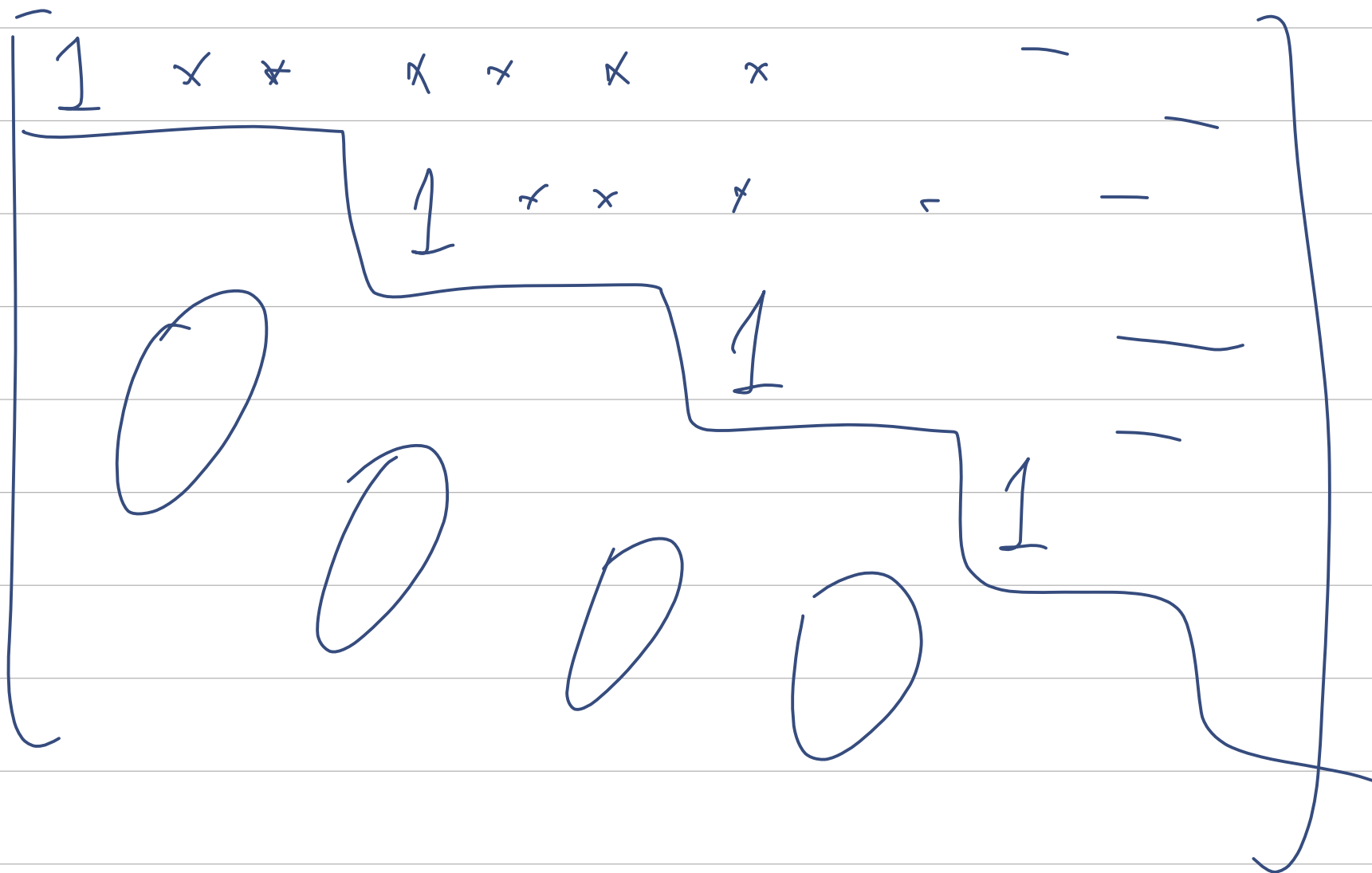
A matrix is in row echelon form (REF) if

- Leading 1s:** The first entry of each non-zero row is 1 (called pivot);
- Leading 1s Moving Right:** If a row has a leading 1, the leading 1 in the next row appears in a later column;
- Zero Rows at the Bottom**

If there is a row whose entries are all zero, then all rows below it have all zero entries.



日期: /



Examples of REF

Examples we've seen before:

Three 5x5 matrices illustrating row echelon form (REF) with red annotations showing row operations:

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Some new examples:

Two 5x5 matrices illustrating row echelon form (REF) with red annotations showing row operations:

$$\begin{bmatrix} 1 & 2 & 2 & 3 & 1 \\ 0 & 1 & 5 & 4 & 5 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

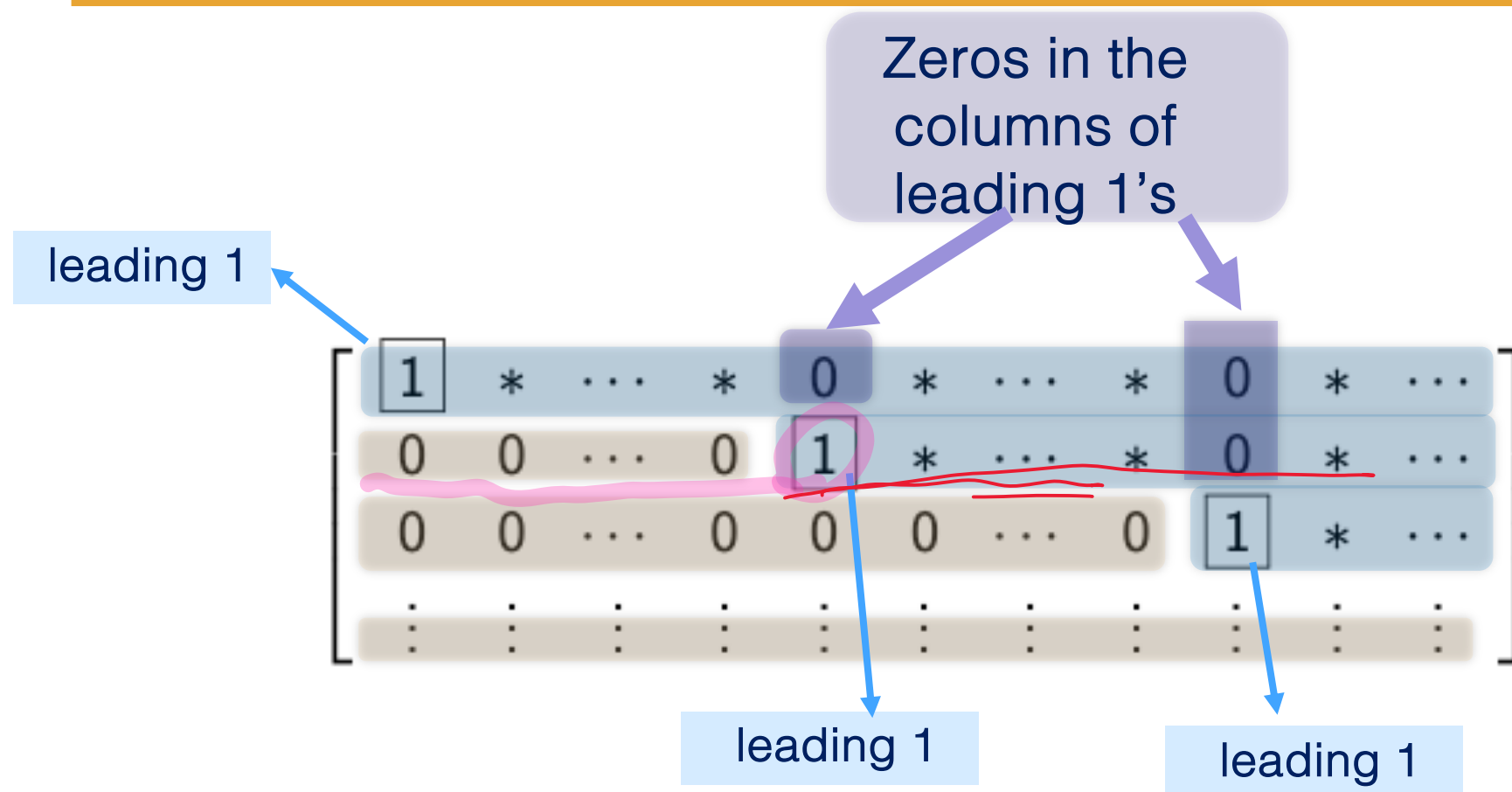
$$\begin{bmatrix} 1 & 2 & 2 & 3 & 1 \\ 0 & 1 & 5 & 4 & 5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

RREF: One More Condition

Definition (Reduced Row-Echelon Form)

A matrix is in reduced row echelon form (RREF) if

- i) **Leading 1s:** The first ^{non-zero} entry of each non-zero row is 1 (called pivot);
- ii) **Leading 1s Moving Right:** If a row has a leading 1, the leading 1 in the next row appears in a later column;
- iii) **Zero Rows at the Bottom**
If there is a row whose entries are all zero, then all rows below it have all zero entries.
- iv) **Zeros in the column of the leading 1:** Each leading 1 is the only nonzero entry in its column (the corresponding column is called **pivot column**)



Examples of RREF

Examples we've seen before:

$$\begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Diagram illustrating a 4x4 matrix in RREF. The first three rows are non-zero and form the identity matrix. The fourth row is all zeros. Red circles highlight the leading ones (1s) in the first three rows. Red arrows point to the first three columns, indicating they are pivot columns.

$$\begin{bmatrix} 1 & 0 & * & 0 & 0 \\ 0 & 1 & * & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Diagram illustrating a 5x5 matrix in RREF. The first three rows are non-zero. The first row has a leading one in column 1. The second row has a leading one in column 2. The third row has a leading one in column 4. Red circles highlight the leading ones. Red arrows point to the first three columns, indicating they are pivot columns.

$$\begin{bmatrix} 1 & 0 & * & 0 & 0 \\ 0 & 1 & * & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Diagram illustrating a 5x5 matrix in RREF. The first three rows are non-zero. The first row has a leading one in column 1. The second row has a leading one in column 2. The third row has a leading one in column 4. The fourth row has a leading one in column 5. Red circles highlight the leading ones. Red arrows point to the first three columns, indicating they are pivot columns.

Other examples:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Diagram illustrating a 5x5 matrix in RREF. The first five rows are non-zero and form the identity matrix. Red circles highlight the leading ones (1s) in the first five rows. Red arrows point to the first five columns, indicating they are pivot columns.

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 5 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Diagram illustrating a 5x5 matrix in RREF. The first three rows are non-zero. The first row has a leading one in column 1. The second row has a leading one in column 2. The third row has a leading one in column 4. Red circles highlight the leading ones. Red arrows point to the first three columns, indicating they are pivot columns.

RREF: Simpler Way to Express

$$\begin{bmatrix}
 \boxed{1} & \boxed{0} & * & \boxed{0} & * \\
 \boxed{0} & \boxed{1} & * & \boxed{0} & * \\
 \boxed{0} & \boxed{0} & 0 & \boxed{1} & * \\
 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0
 \end{bmatrix}
 \longrightarrow
 \begin{bmatrix}
 \boxed{1} & \boxed{0} & \boxed{0} & * & * \\
 \boxed{0} & \boxed{1} & \boxed{0} & * & * \\
 \boxed{0} & \boxed{0} & \boxed{1} & 0 & * \\
 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0
 \end{bmatrix}$$

Property:

If a matrix A is in reduced row echelon form (RREF), then exchanging columns of A can lead to the form

$$\begin{bmatrix}
 I_k & F \\
 0 & 0
 \end{bmatrix}$$

where I_k is the $k \times k$ identity matrix.

changing column not used in $GJ-E$. This Prop. is for understanding

Exercise of RREF

Are these RREF?

$$\begin{bmatrix} 1 & 0 & * & 0 & * \\ 0 & 1 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

X

→ if one row is all-zero every thing below shall be 0.

$$\begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

X

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

X

$$\begin{bmatrix} 0 & 1 & 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

✓

→ fine
→ not pivot

End of Each Stage of GJ-E for Square System

Proposition 7.1 If we solve an $n \times n$ linear system of equations by Gauss-Jordan Elimination, then we will:
--obtain REF (which is upper triangular) in the process,
--obtain RREF in the end.

Proof Sketch:

Lec 4 said:

If at each iteration of the forward elimination, the next diagonal entry is nonzero
(i.e. obtain n pivots without row exchange)
then we obtain a unique solution.

What if the next candidate (on the bottom right of the k -th pivot) is zero?

Row exchange

What if row exchange cannot create a non-zero entry?

Go to next column to search for nonzero entries. [see New Case 1&2 before]

What if cannot find nonzero entries after searching all columns?

This means everything below the operated row is zero → obtained REF.

At the end of forward elimination, we must obtain an REF.

From REF, eliminating all entries above all pivots, we obtain an RREF.



Summary Today (write Your Own)

One sentence summary:

Detailed summary:

Quiz 1 Instruction

Time: 20-30 mins. During Tutorial.

Closed book/notes; no electronic devices.

Write your name and ID.

Write solutions directly on the quiz sheet.

Use the blank paper to cover quiz sheets, until announcing start.